

Examples of Figures for NIST SP 800-90B non-IID track

May 9, 2026

1 Finding the minimum of estimators listed in §6.3.x as per §6.2 of NIST SP 800-90B

Figures 1 and 2 show the min-entropy estimates using §6.3.x of NIST SP 800-90B for truerand_1bit.bin [3] and for rand1_short.bin [5], respectively.

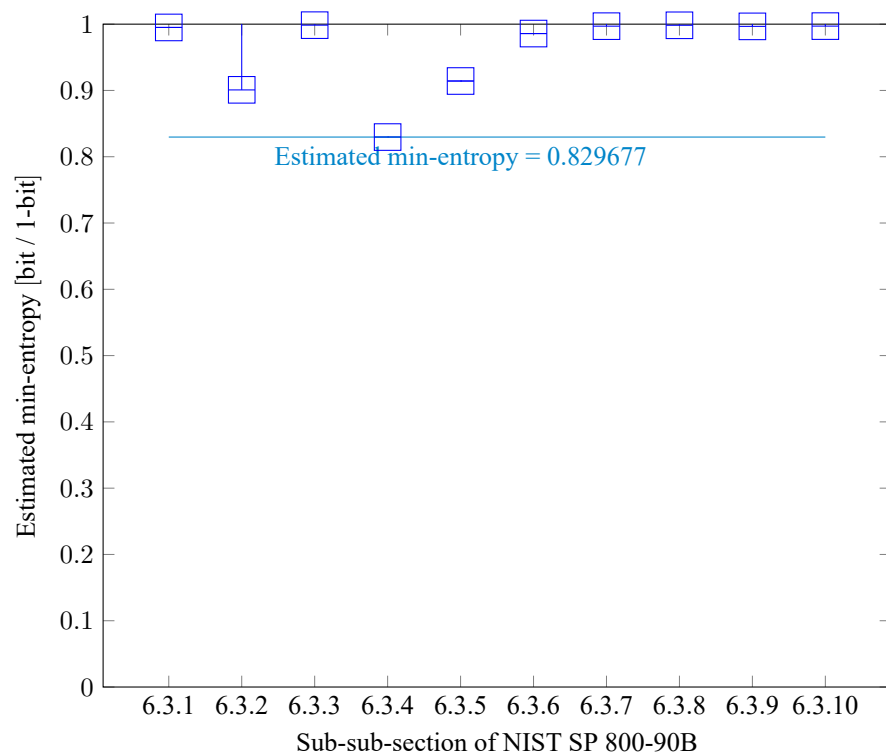


Fig. 1 Estimated Min-Entropy using §6.3 of NIST SP 800-90B for truerand_1bit.bin [3]

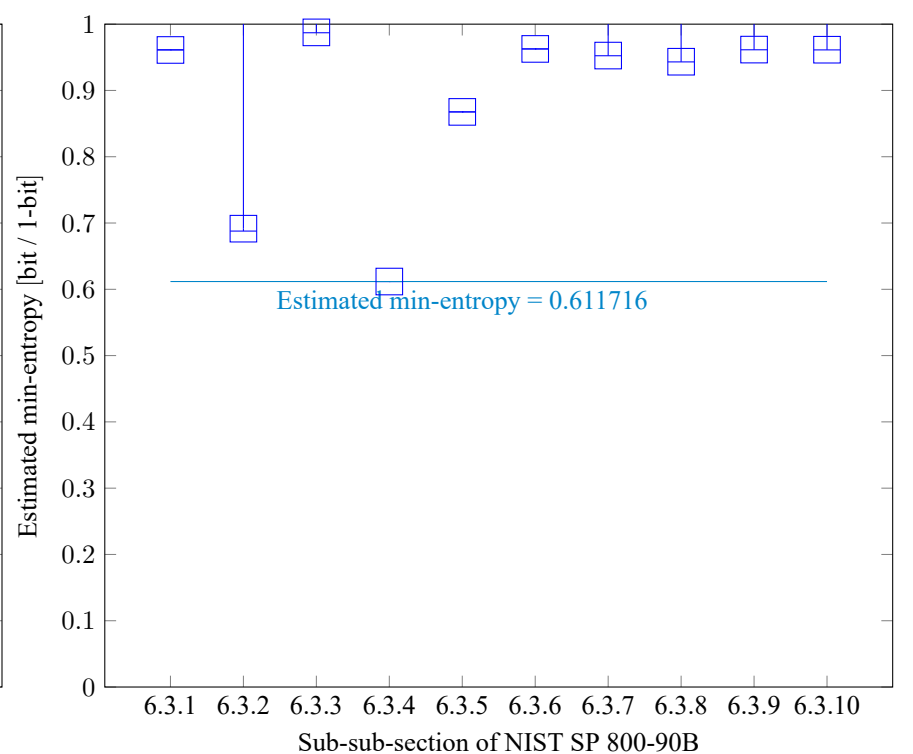


Fig. 2 Estimated Min-Entropy using §6.3 of NIST SP 800-90B for rand1_short.bin [5]

2 Examples of figures obtained by the Most Common Value Estimate

Figures 3 and 4 show the distribution of x_i for truerand_8bit.bin [4] and for normal.bin [7], respectively. Nearly 'white' distribution can be seen in Figure 3, while a distribution similar to normal distribution can be seen in Figure 4.

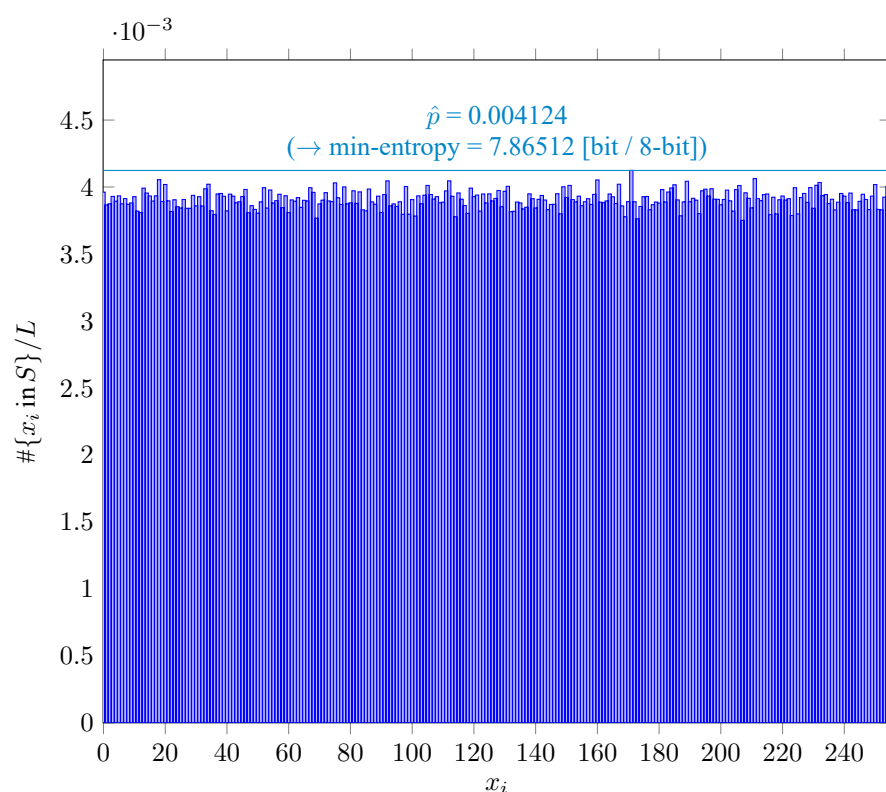


Fig. 3 Distribution of x_i for truerand_8bit.bin [4]

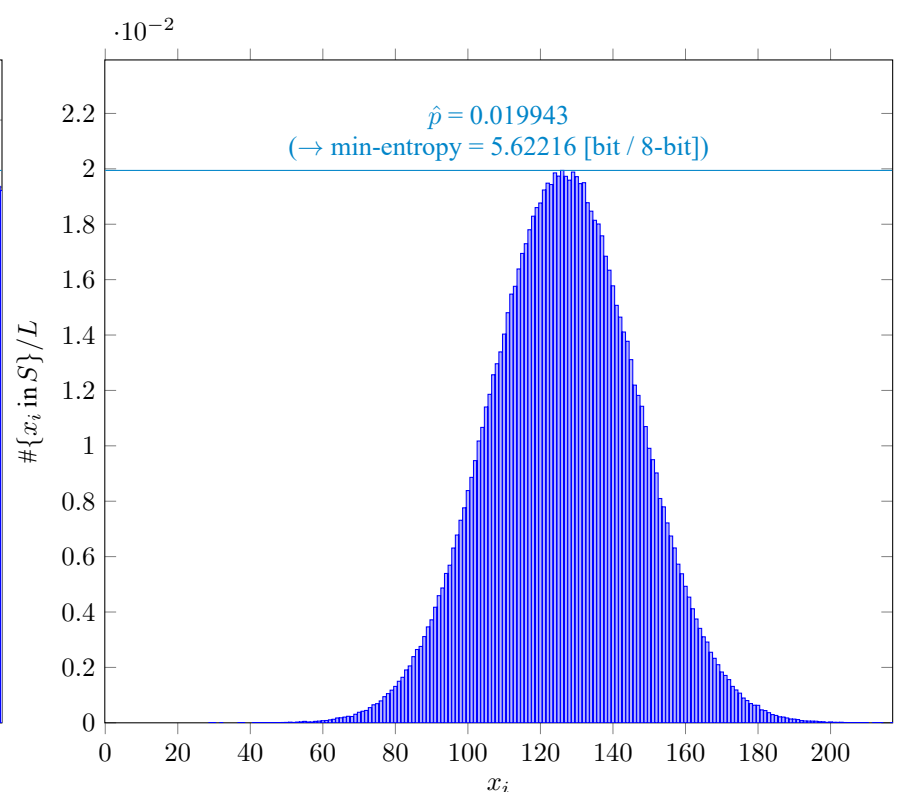


Fig. 4 Distribution of x_i for normal.bin [7]

3 Example of figure obtained by the Collision Estimate

Figure 5 shows visually how to solve the equation in step 7 by using the following:

1. $\bar{X}'$ on the l.h.s. of equation in step 7 of the Collision Estimate,
2. $g(p)$ as the r.h.s. of equation in step 7 of the Collision Estimate, and
3. solution of p with respect to given X' .

As shown in ref.[2], the r.h.s. of equation in step 7 of the Collision Estimate can be rewritten to as follows:

$$\begin{aligned} g(p) &\equiv pq^{-2} \left[1 + \frac{1}{2}(p^{-1} - q^{-1}) \right] F(q) - pq^{-1} \frac{1}{2}(p^{-1} - q^{-1}) \\ &= 2[p(1-p) + 1] \\ &= 2 \left[-\left(p - \frac{1}{2}\right)^2 + \frac{5}{4} \right] \end{aligned} \quad (1)$$

Here we used the following property:

$$F(q) = q(2q^2 + 2q + 1) \quad (2)$$

Now finding the solution of p with respect to given X' , reduces to solving the following quadratic equation:

$$\bar{X}' = 2[p(1-p) + 1]. \quad (3)$$

If $\bar{X}'$ is in range $[2, \frac{5}{2}]$, the solution p can be expressed by the following equation:

$$p = \frac{1}{2} + \sqrt{\frac{5}{4} - \frac{\bar{X}'}{2}}. \quad (4)$$

In Figure 5, we get the solution p is 0.673833 for given $X' = 2.43956$.

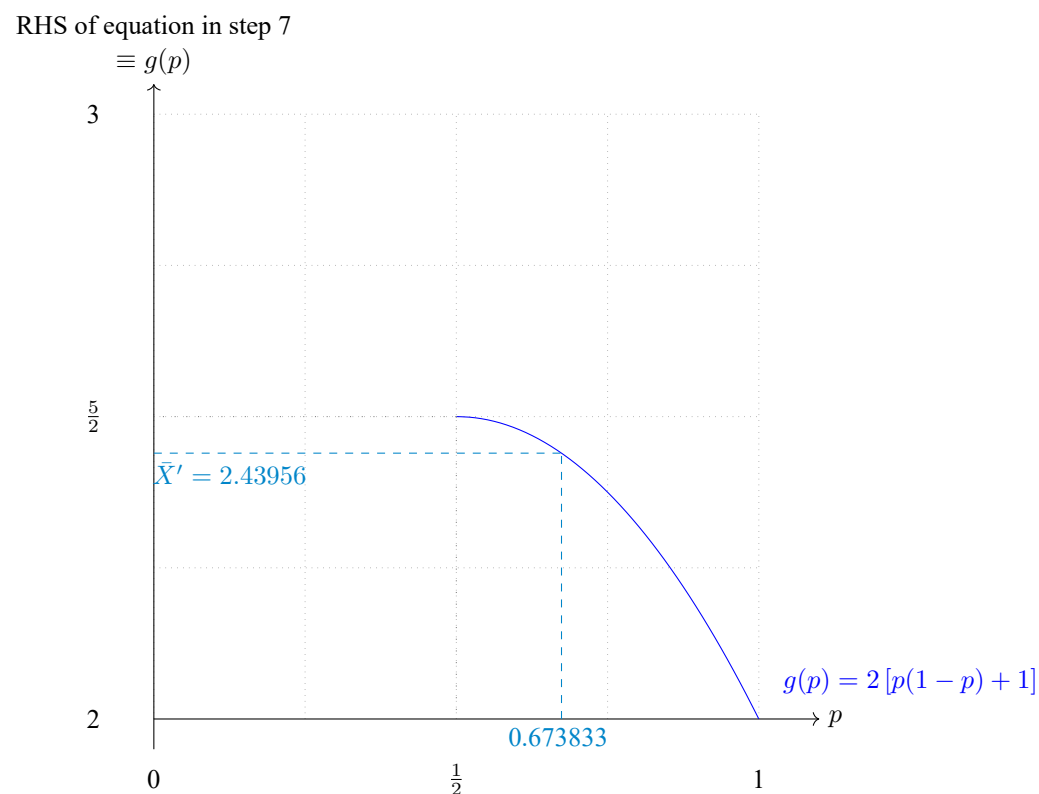


Fig. 5 Solution to the equation in step 7 in §6.3.2 of NIST SP 800-90B for data.pi

4 Examples of figures obtained by the Markov Estimate

Figures 6 and 7 show the estimated min-entropy for truerand_1bit.bin [3] and for normal.bin [6], respectively. In both figures, sequence indices 2, 3, 12, and 13 are added in addition to sequences identified in NIST SP 800-90B[1]. This addition is reasonable because min-entropy estimates based on these sequences are comparable to those of sequences identified in NIST SP 800-90B, and sometimes very close to the minimum, therefore should not be neglected.

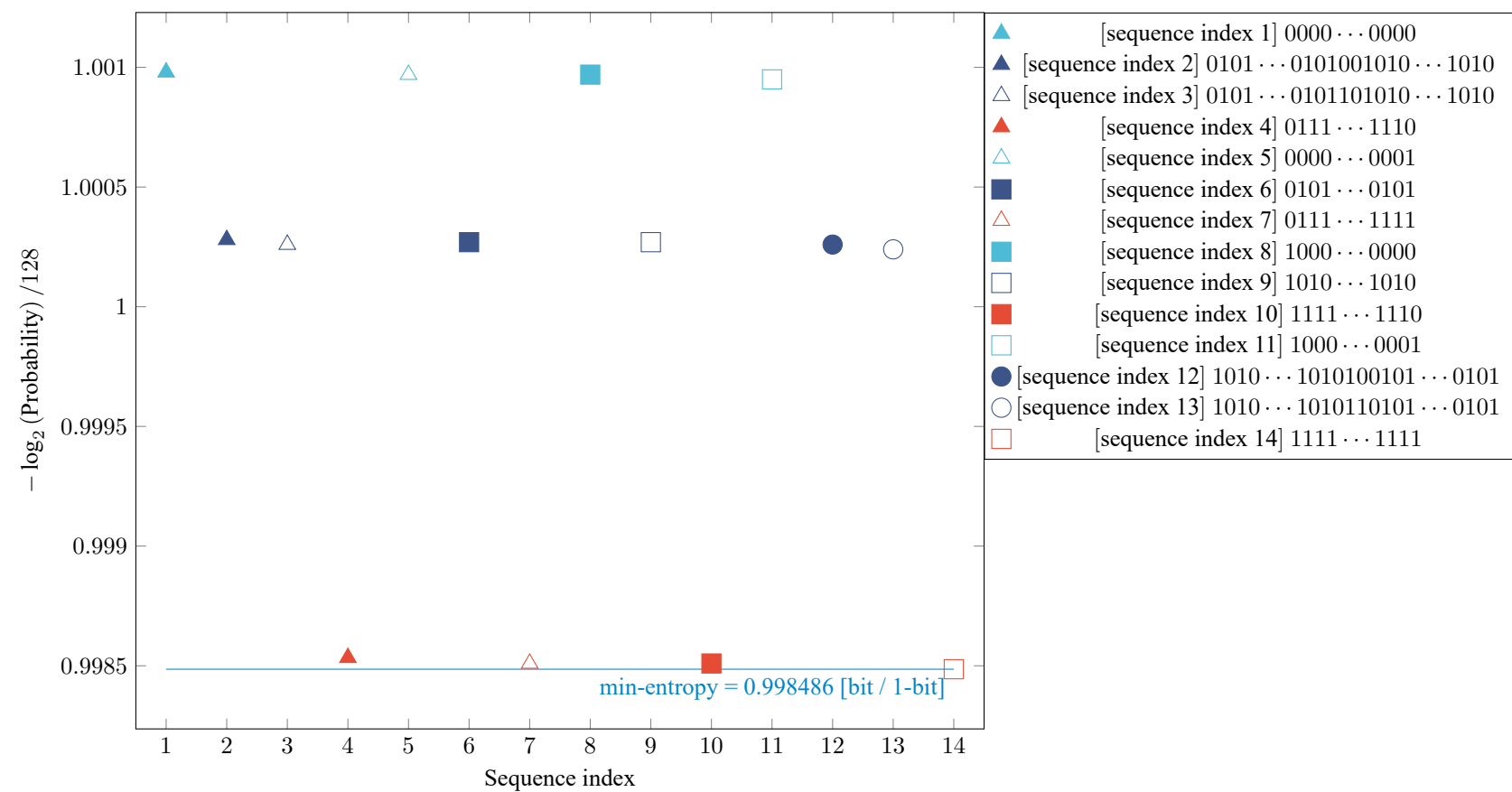


Fig. 6 Estimated Min-Entropy using §6.3.3 of NIST SP 800-90B

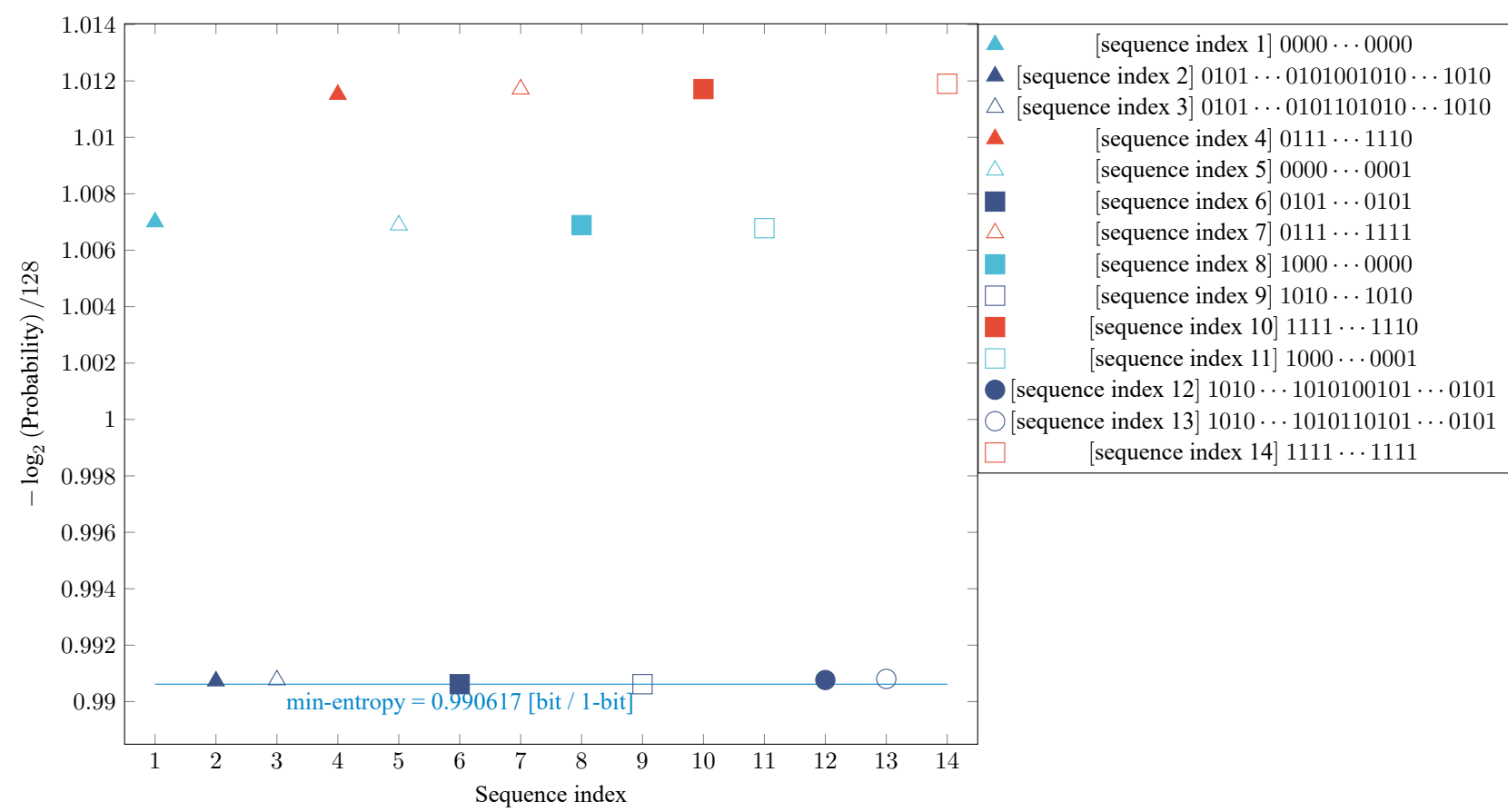


Fig. 7 Estimated Min-Entropy using §6.3.3 of NIST SP 800-90B

5 Example of figures obtained by the Compression Estimate

Figure 8 shows the distribution of intermediate value D_i in step 5 of the Compression Estimate for truerand_1bit.bin [3]. Here, a semi-logarithmic plot is used, and the vertical green bar shows the sample mean, $\bar{X}$.

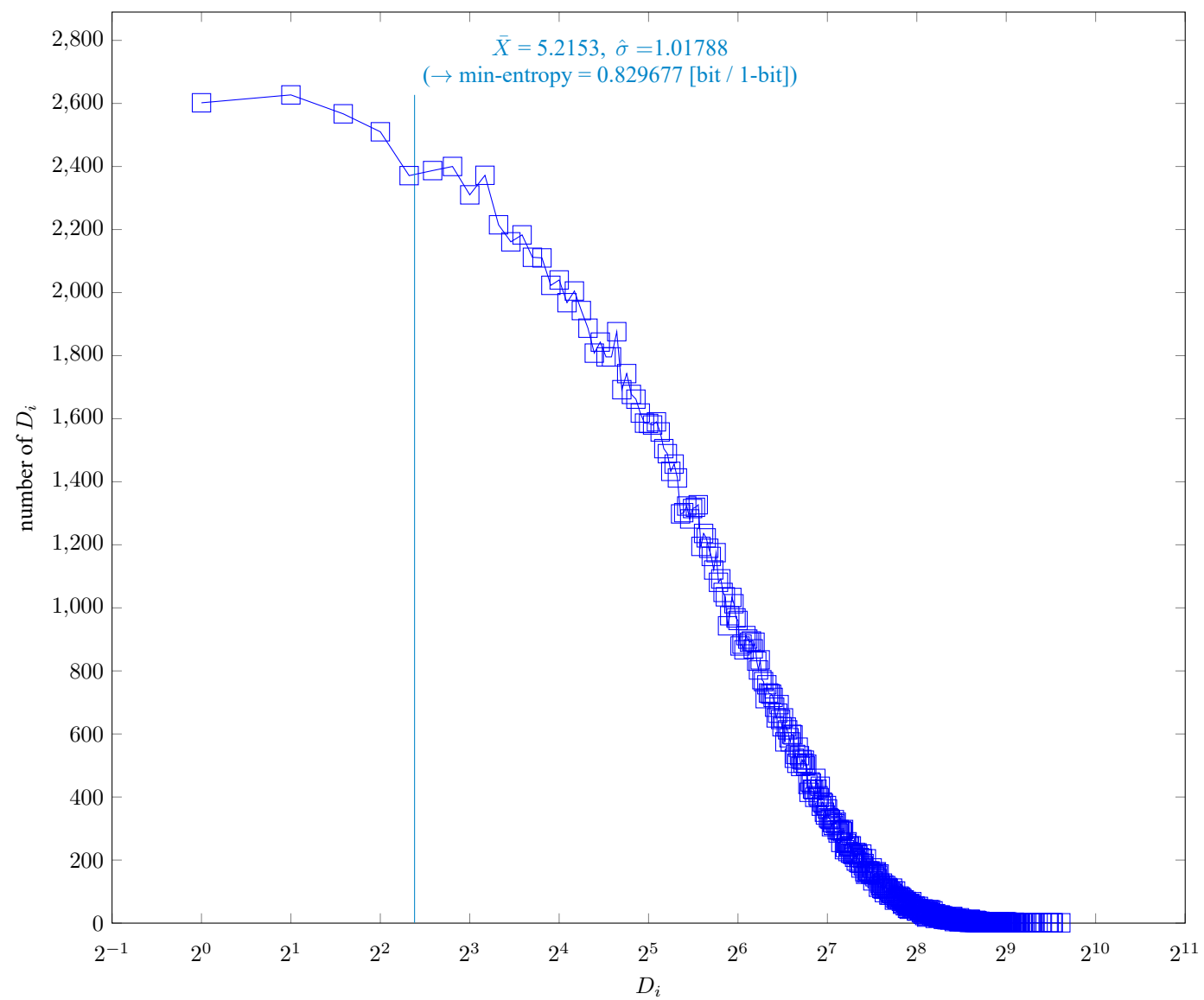


Fig. 8 Distribution of intermediate value D_i

6 Example of figure obtained by the t -Tuple Estimate

Figure 9 shows the distribution of intermediate value $Q[i]$ in step 2 of the t -tuple Estimate for truerand_1bit.bin [3]. This figure exhibits a monotonically decreasing behaviour, which would be expected intuitively.

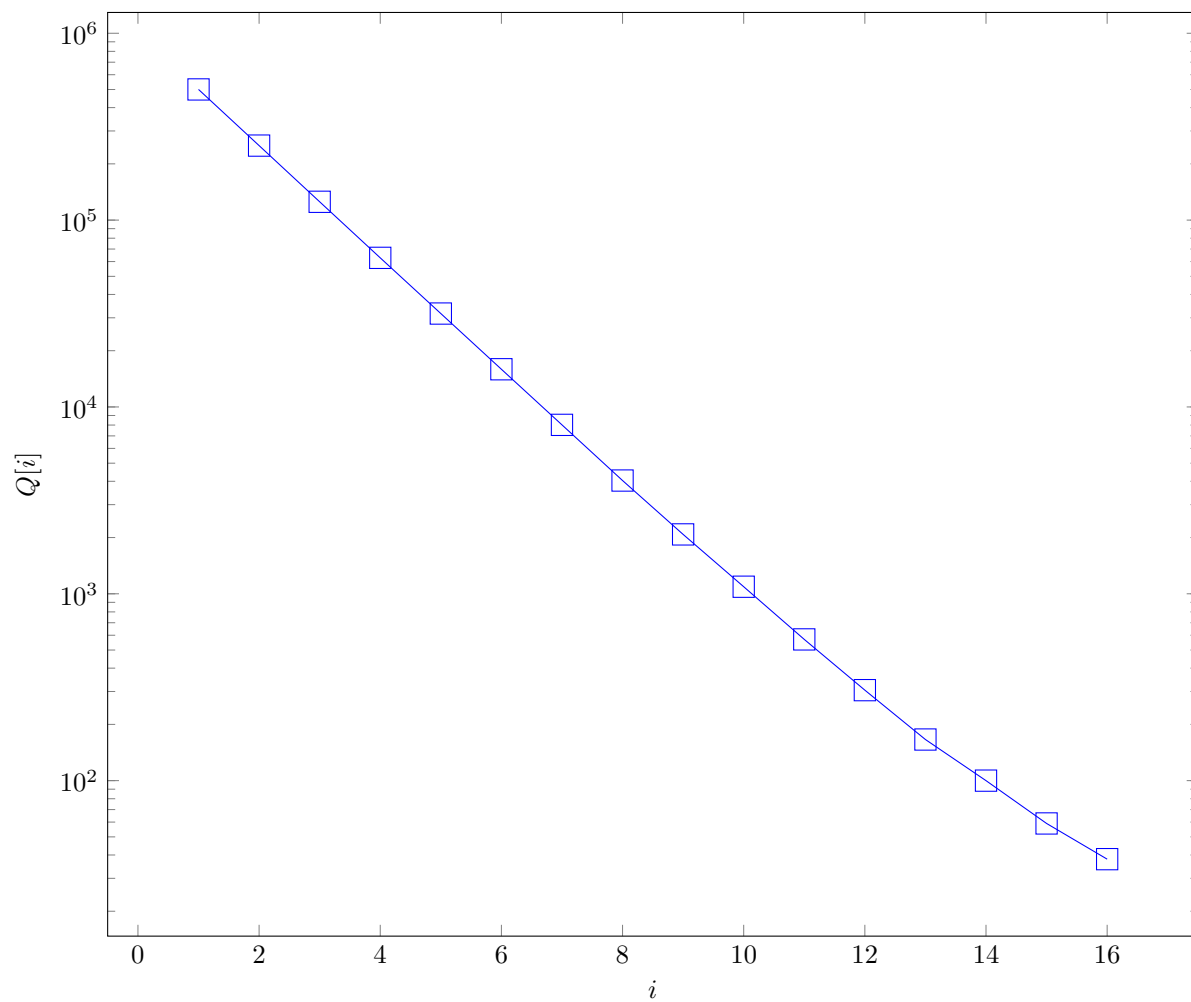


Fig. 9 Intermediate value $Q[i]$ in §6.3.5 of NIST SP 800-90B for truerand_1bit.bin [3]

7 Example of figure obtained by the Longest Repeated Substring (LRS) Estimate

Figure 10 shows the multiplicity of W -tuple having the same C_i value in step 3 of the Longest Repeated Substring (LRS) Estimate for truerand_1bit.bin [3].

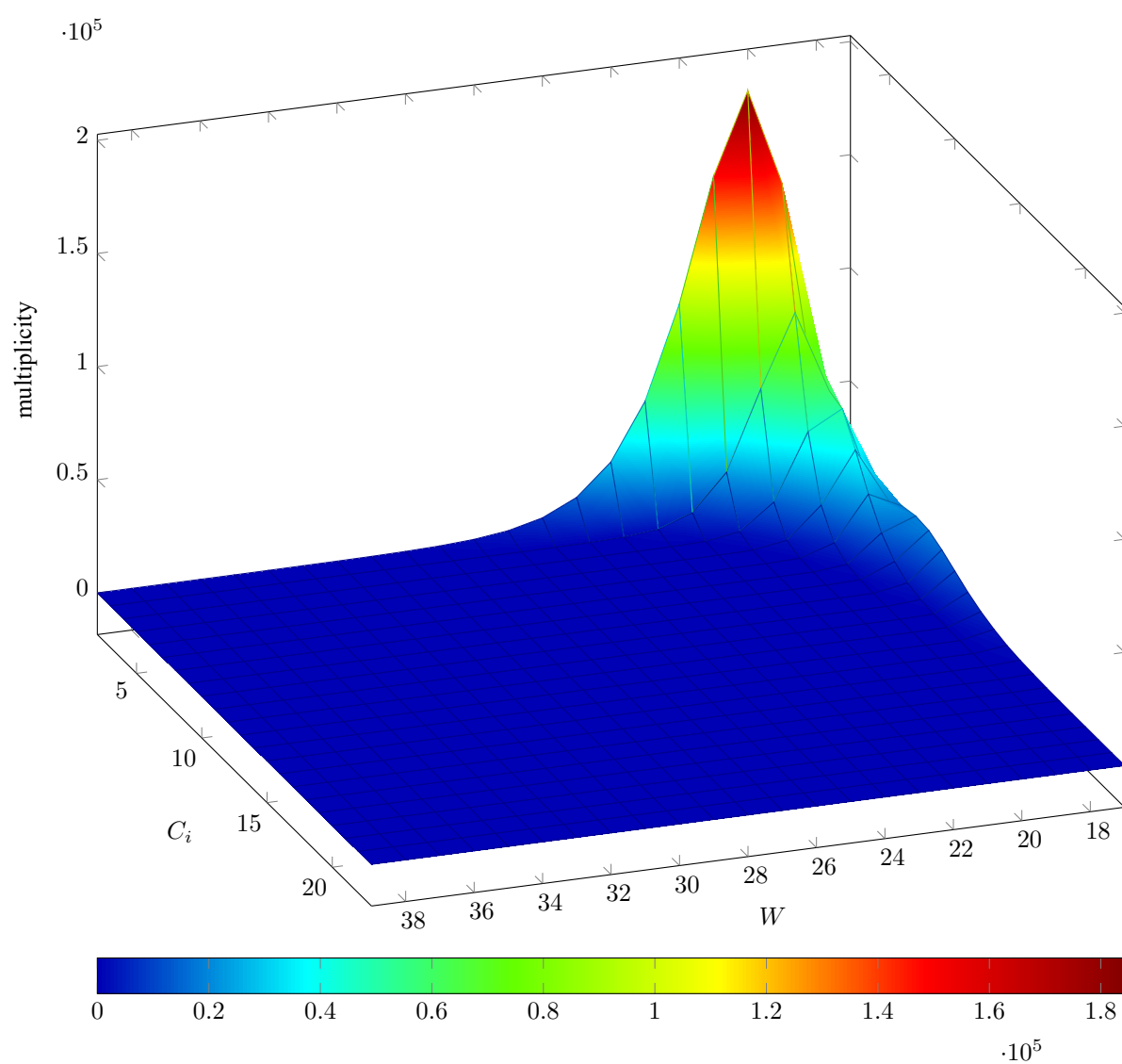


Fig. 10 Distribution of the multiplicity of W -tuple having the same C_i value in Step 3 of §6.3.6 of NIST SP 800-90B for truerand_1bit.bin [3]

8 References

- [1] Meltem Sönmez Turan, Elaine Barker, John Kelsey, Kerry A. McKay, Mary L. Baish, Mike Boyle, *Recommendation for the Entropy Sources Used for Random Bit Generation*, NIST Special Publication 800-90B, Jan. 2018 <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-90B.pdf>.
- [2] G. Sakurai, *Implementation Notes for entropy estimation based on NIST SP800-90B non-IID track*, Sep. 2025 https://github.com/g-g-sakura/AnotherEntropyEstimationTool/blob/main/documentation/SP800-90B_EntropyEstimate_ImplementationNotes.pdf.
- [3] Data file named *truerand_1bit.bin*, https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/truerand_1bit.bin.
- [4] Data file named *truerand_8bit.bin*, https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/truerand_8bit.bin.
- [5] Data file named *rand1_short.bin*, https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/rand1_short.bin.
- [6] Data file named *rand4_short.bin*, https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/rand4_short.bin.
- [7] Data file named *normal.bin*, https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/normal.bin.
- [8] Data file named *data.pi.bin*, https://github.com/usnistgov/SP800-90B_EntropyAssessment/blob/master/bin/data.pi.bin.